Tutorial 10: Oct 8, 2025

(1) Compute $\displaystyle\int_C (y\,dx + z\,dy + x\,dz)$

C is the intersection of $x^2+y^2+z^2 = a^2$ and the plane $x+y+z=0$. (Apostol, vol 2, p 442)

(2) Compute $\nabla \cdot \left(\dfrac{\vec{r}}{r^3}\right)$; $\vec{r} = xi + yj + zk$.

$\nabla \times \left(\dfrac{\vec{r}}{r^3}\right)$. (A great deal of trouble can be avoided by ~~regag~~ recognizing $\dfrac{\vec{r}}{r^3}$ as something being familiar)

(3) Find ϕ such that $\nabla\phi = \vec{F}$ in the following two cases: $\vec{F} = (e^x \cos y + yz)\, i + (zx - e^x \sin y)\, j + (xy+z)\, k$

$$\vec{F} = \frac{2x\, i}{(x^2+y^2)^2} + \frac{2y}{(x^2+y^2)^2}\, j$$

$\vec{F} = \dfrac{yi - xj}{x^2+y^2}$ in the domain $\{(x,y) / x>0, y>0\}$

(4) Compute $\displaystyle\iiint_E \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right) dx\,dy\,dz$

Where $E = \left\{ (x,y,z) / \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} < 1 \right\}$.

Compute $\displaystyle\iiint_S \frac{dx\,dy\,dz}{\left((x-a)^2+(y-b)^2+(z-c)^2\right)^{\frac{1}{2}}}$; S is the ~~a~~ ball of radius R centre origin. (a,b,c) lies outside

(5) Disiuss problem on Spherical Shell with zero thickness on pages 218-219. Can you see any relevance of this with the famous problem on Simple harmonic motion: A tunnel is made ~~akan~~ from north pole to South pole of Earth. A parcel is dropped from north pole. When does it reach South pole ....

(6) Do problems (ii) - (iv) on page 222 of notes.

(7) p 217. Do area of a surface of revolution - use the parametrization you discussed in tutorial 9.

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